

CASE STUDY

Tesco Grocery 1.0 Dataset vs. Income

What is

Tesco Grocery 1.0

dataset?

Tesco Grocery 1.0 Dataset



RESEARCHERS



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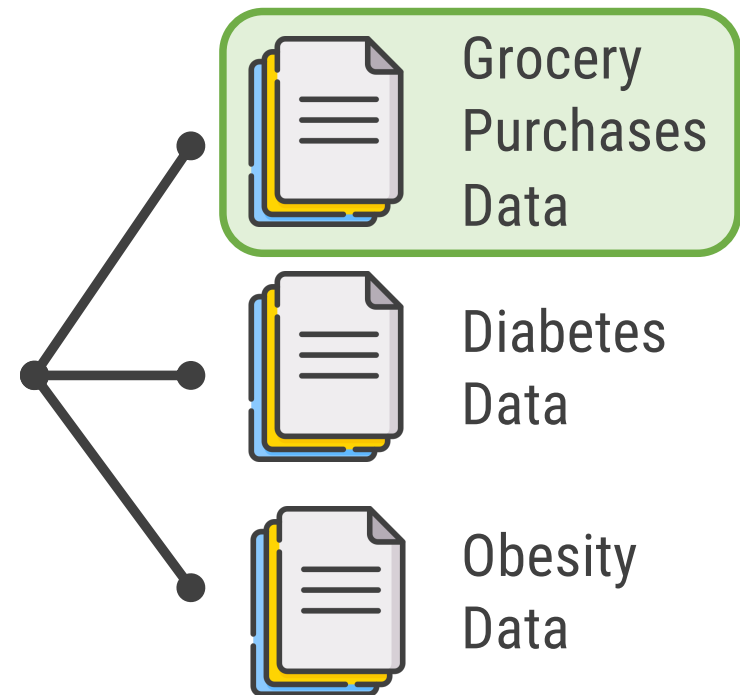


Rossano Schifanella

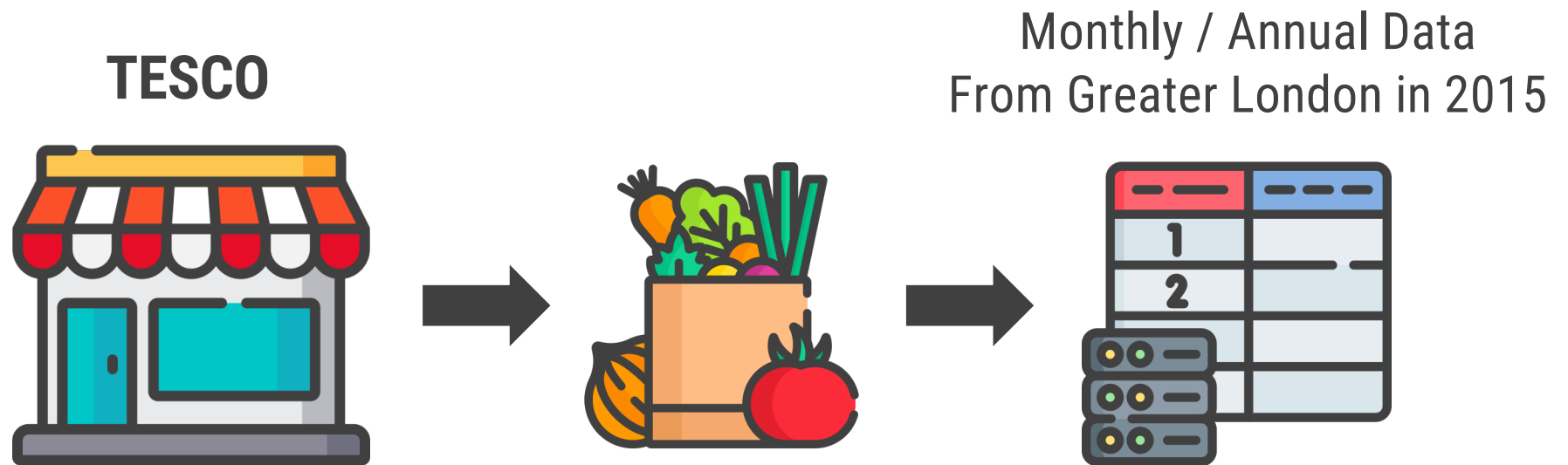


Lucia Del Prete

Tesco Grocery 1.0 Dataset



Tesco Grocery 1.0 Dataset



Tesco Grocery 1.0 Dataset

TESCO

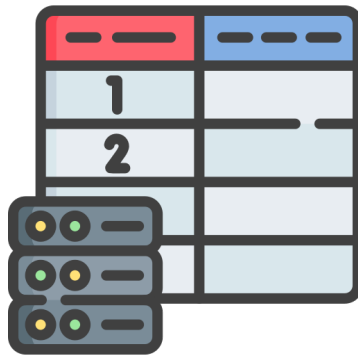


IN 2015

28% of the market share

9th highest-grossing retailer

Tesco Grocery 1.0 Dataset



1.6M Clubcard holders

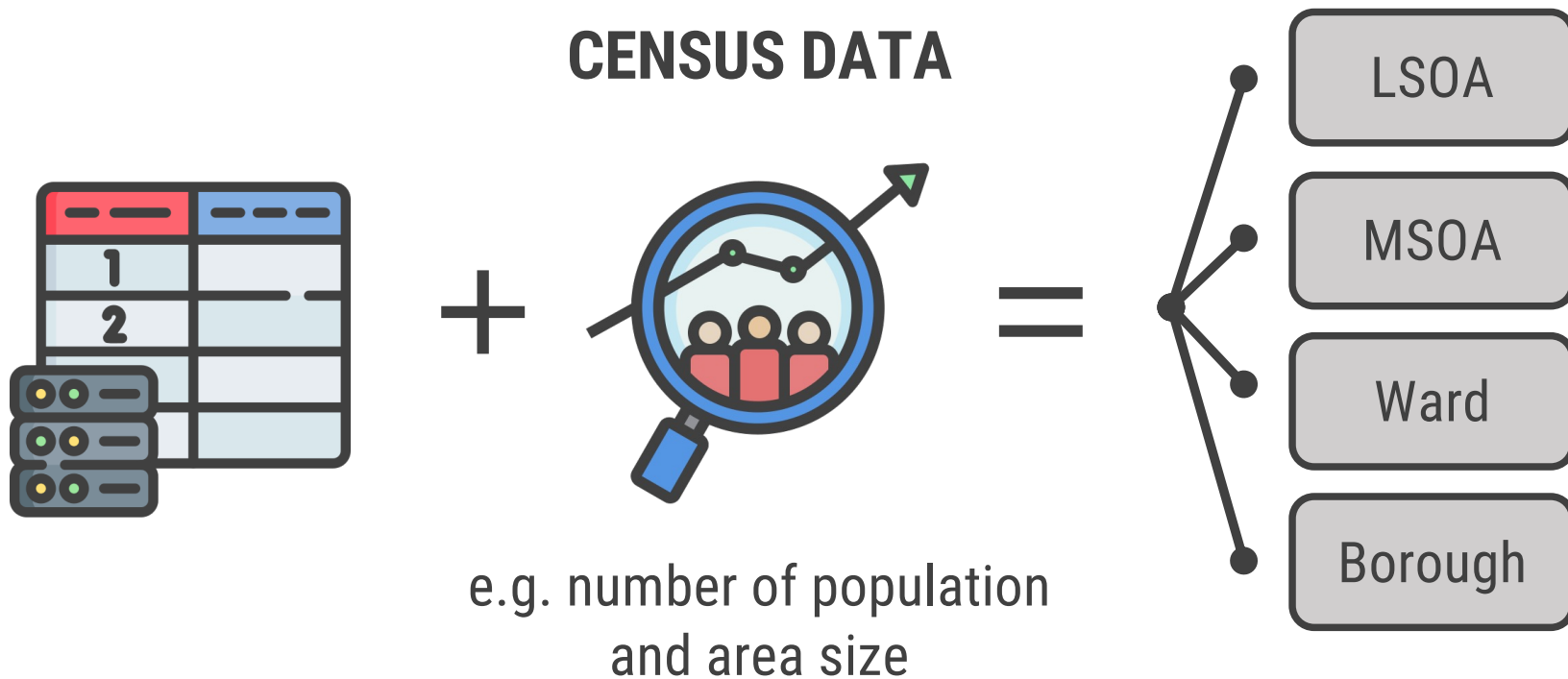
411 Tesco stores in Greater London

From Jan 1st to Dec 31st in **2015**

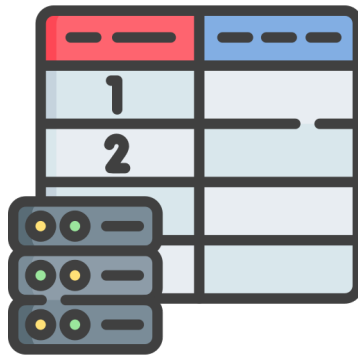
420M purchased items

67K distinct goods

Tesco Grocery 1.0 Dataset



Tesco Grocery 1.0 Dataset



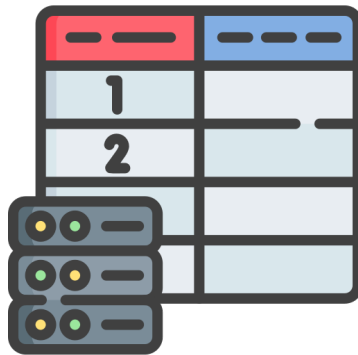
Calculated products consumed by each area

- Weight, Volume, Nutrition and Product categories

Represented in statistics

- Average, Percentile, Standard deviation, Confidence interval, Probability distribution and Entropy

Tesco Grocery 1.0 Dataset



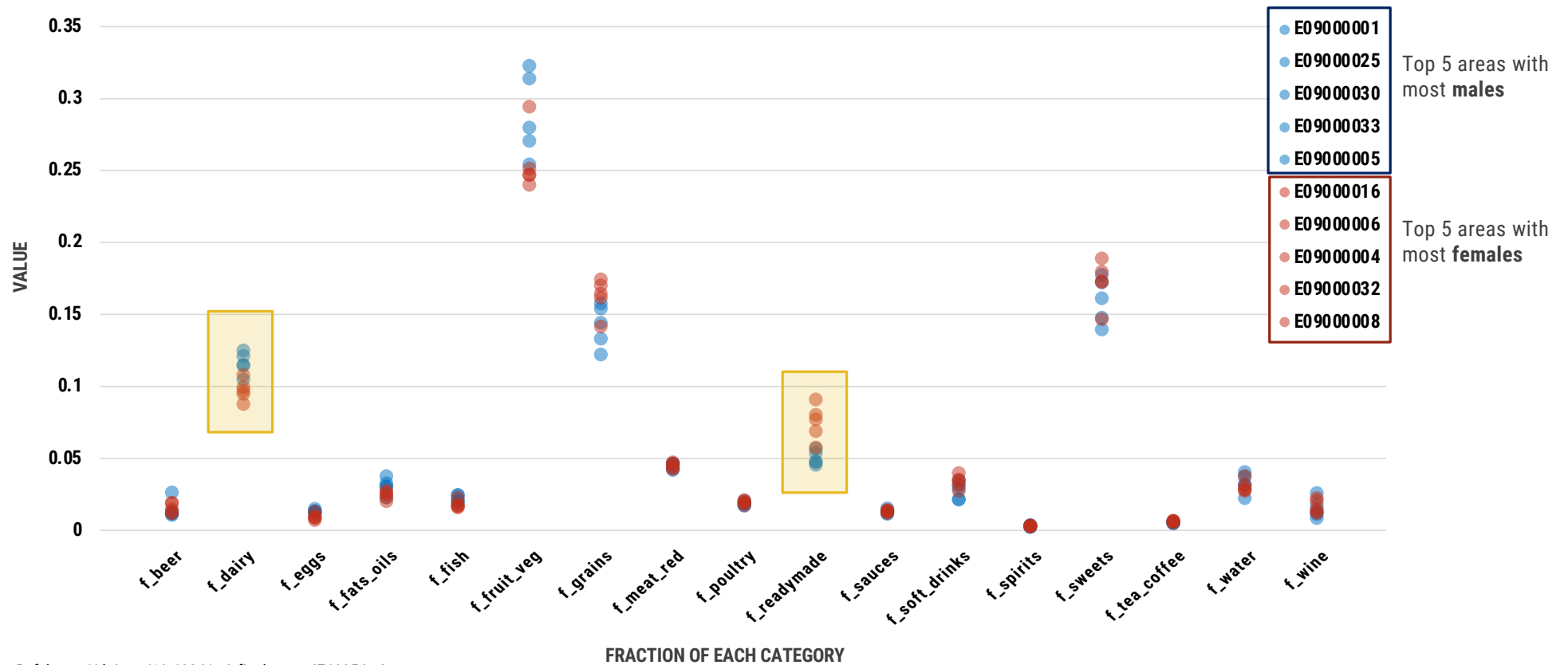
Biases and Limitations

- Only Clubcard holders
- Only from in-store purchases
- Not equal in penetration rate in each area
- Not identify how many people each purchase is for
- Not reflect purchases in other stores or restaurants

What **insights**
we could find from
Tesco Grocery 1.0 dataset?

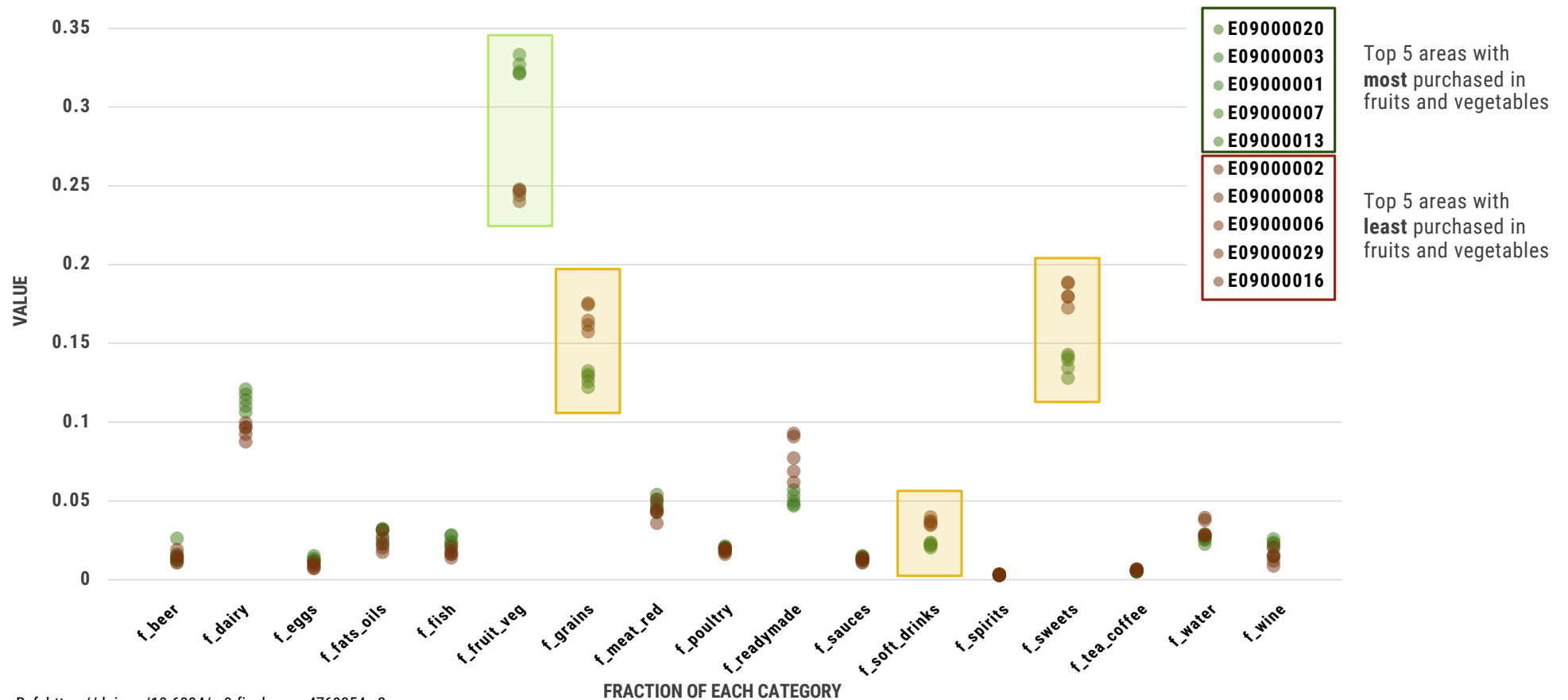
INSIGHT 1

What are the differences in grocery purchases between areas with more males and females?



INSIGHT 2

What are the differences in grocery purchases between areas with high and low purchases of fruits and vegetables?



What is
Income dataset?

Income Dataset

The screenshot shows the Office for National Statistics website. The header includes the logo and name 'Office for National Statistics', a language selector for 'English (EN)' and 'Cymraeg (CY)', and a navigation menu with links: 'Release calendar', 'Methodology', 'Media', 'About', and 'Blog'. Below the header is a horizontal navigation bar with categories: 'Home', 'Business, industry and trade', 'Economy', 'Employment and labour market', 'People, population and community', and 'Taking part in a survey?'. A search bar is located below the navigation bar with the placeholder text 'Search for a keyword(s) or time series ID' and a green search icon. The main content area shows a breadcrumb trail: 'Home > Economy > Regional accounts > Gross disposable household income'. Below the breadcrumb trail, the word 'Dataset' is displayed, followed by the title 'Regional gross disposable household income: local authorities by ITL1 region'.

RELEASED DATE

13 OCTOBER 2022

Ref: <https://www.ons.gov.uk/economy/regionalaccounts/grossdisposablehouseholdincome/datasets/regionalgrossdisposablehouseholdincomelocalauthoritiesbyitl1region>

Income Dataset



Contains London **gross income** per head

Data from **1997** to **2020**

Separated at **Borough** level

What **insight**
we could find from
income dataset?

INSIGHT: Gross disposable household income (£) at Borough level over the years

Area code	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
E09000001	90,955	100,333	93,408	99,183	97,691	99,214	104,460	112,215	126,070	135,758	145,091	160,358	158,242	152,611	148,259	179,835	218,893	226,637	227,911	217,301	214,667	208,106	179,272	147,813
E09000002	8,842	9,089	9,510	9,999	10,657	11,049	11,713	12,311	12,743	13,093	13,837	14,202	14,169	14,275	14,348	15,106	15,353	15,565	16,270	16,625	17,003	18,148	18,832	19,265
E09000003	12,615	13,265	13,976	15,155	15,672	16,163	16,923	17,522	18,604	19,380	20,670	21,276	21,004	21,445	21,570	23,076	24,030	25,271	28,080	28,717	29,068	30,624	31,136	30,460
E09000004	11,663	11,776	12,247	12,888	13,708	14,012	14,535	15,034	15,519	15,964	16,917	17,375	17,488	17,874	17,945	18,870	19,418	19,881	21,207	21,760	21,897	23,001	23,790	23,758
E09000005	10,784	11,226	12,065	12,904	13,436	13,966	14,638	15,165	16,187	16,561	17,585	18,011	18,163	18,349	18,501	19,853	21,043	21,911	23,510	23,724	24,089	25,650	26,226	26,438
E09000006	13,808	14,431	15,174	16,289	16,824	17,136	18,067	19,123	20,010	20,606	21,773	22,021	21,707	22,229	22,622	23,912	24,614	25,618	27,424	27,690	27,734	29,385	30,025	29,854
E09000007	16,075	17,822	19,004	21,065	20,960	20,976	21,301	23,045	25,558	28,076	31,379	34,307	34,190	33,086	31,706	34,063	37,391	37,995	40,397	40,774	41,510	44,243	44,009	41,810
E09000008	10,174	10,553	11,344	12,346	12,946	13,386	14,142	14,948	15,757	16,266	17,291	17,632	17,401	17,756	17,981	19,012	19,657	19,959	21,498	21,855	22,254	23,846	24,676	24,636
E09000009	12,886	13,217	13,953	14,767	15,324	15,695	16,421	16,926	17,641	18,103	19,170	19,677	19,629	20,085	20,258	21,674	23,008	24,276	26,253	26,354	26,409	27,477	28,128	28,337
E09000010	10,853	11,228	11,604	12,564	13,313	13,446	13,996	14,677	15,288	15,746	16,922	17,443	17,360	17,557	17,558	18,708	19,147	19,849	21,461	21,969	22,037	23,094	24,105	24,188
E09000011	13,728	13,786	14,438	15,197	15,751	15,841	16,258	16,726	17,185	17,508	18,443	18,817	18,700	18,870	18,748	19,317	20,021	20,565	21,621	22,039	22,270	23,384	24,424	24,761
E09000012	7,273	7,980	8,818	9,987	10,573	10,910	11,343	12,214	13,339	14,484	15,809	16,306	16,698	17,208	17,549	18,506	19,936	20,742	22,507	23,203	23,437	24,670	25,852	25,952
E09000013	21,077	21,234	22,403	24,194	24,179	23,595	24,404	26,060	28,114	30,242	33,157	34,675	34,495	33,777	32,539	34,811	38,925	41,268	42,654	41,819	42,608	44,109	43,730	43,998
E09000014	13,974	14,243	15,164	16,162	16,501	16,715	17,062	17,985	19,043	20,075	21,445	21,538	21,508	22,112	22,396	23,820	24,325	25,370	27,028	26,523	27,625	29,024	30,049	30,099
E09000015	12,769	13,473	14,213	15,241	15,879	16,346	17,059	17,495	18,285	18,607	19,947	20,487	19,872	20,350	20,513	21,905	23,193	23,970	25,845	26,406	26,808	28,292	28,374	27,854
E09000016	11,207	11,662	12,107	13,106	13,892	14,193	14,727	15,219	15,754	16,152	17,327	17,881	17,704	18,189	18,337	19,407	20,028	20,262	21,725	22,298	22,551	23,838	24,274	24,055
E09000017	10,279	10,764	11,614	12,429	13,150	13,628	14,380	15,008	15,783	16,196	17,126	17,590	17,543	17,944	17,894	18,873	19,770	20,016	21,485	21,921	22,063	22,953	23,688	23,621
E09000018	14,207	14,442	15,128	15,911	16,443	16,757	17,541	17,984	18,546	18,926	19,562	19,908	19,753	19,760	19,608	20,855	21,769	22,403	24,143	23,758	24,139	25,684	26,684	26,951
E09000019	16,617	16,769	18,299	19,601	20,003	20,416	20,739	21,779	22,933	24,543	26,333	26,968	26,710	27,283	27,059	28,345	28,811	29,585	31,199	30,669	30,741	32,300	33,261	32,756
E09000020	34,846	34,560	35,735	38,233	37,352	36,662	37,258	39,941	42,827	47,726	54,450	59,110	58,835	56,774	56,104	62,152	68,981	74,087	77,467	76,952	78,689	84,897	82,283	79,325
E09000021	14,471	14,670	15,409	16,489	16,803	16,995	17,665	18,440	19,157	19,603	20,899	21,203	20,895	21,269	21,409	22,606	23,436	23,735	25,073	25,396	25,309	27,550	27,532	26,963
E09000022	11,041	11,344	12,339	13,477	13,739	13,972	14,418	15,228	16,249	17,291	18,609	18,986	19,053	19,679	19,820	20,920	21,864	22,836	25,844	27,077	27,151	29,092	30,035	30,386
E09000023	8,727	9,049	9,830	10,707	11,151	11,441	12,089	12,873	13,854	14,841	15,858	16,122	16,338	16,990	17,255	18,286	19,050	19,937	21,329	21,718	22,463	23,652	24,762	24,991
E09000024	15,972	16,388	17,205	18,294	18,622	19,004	19,917	21,187	22,134	22,634	23,999	24,472	23,871	24,694	25,009	26,347	27,720	28,635	30,517	30,954	31,024	33,008	34,422	34,102
E09000025	5,561	6,083	6,839	7,839	8,400	8,704	9,436	10,297	11,515	12,534	13,295	13,479	13,590	13,569	13,552	14,210	14,872	15,148	16,462	17,034	17,813	18,996	19,946	20,102
E09000026	11,347	11,580	12,043	12,933	13,641	13,941	14,319	14,889	15,463	15,760	16,697	17,079	16,896	17,190	17,148	18,247	18,639	19,078	20,826	21,119	21,904	23,508	23,808	23,865
E09000027	21,361	21,769	22,639	24,257	24,618	24,775	25,295	25,942	27,037	28,168	30,333	31,355	31,175	31,631	31,752	33,906	35,578	36,885	39,980	39,187	39,495	42,504	43,400	43,037
E09000028	11,155	11,605	12,455	13,532	13,813	14,060	14,703	15,538	16,600	17,538	18,754	19,140	19,138	19,759	19,924	20,975	21,886	23,090	24,677	25,048	25,585	27,540	28,014	27,878
E09000029	12,041	12,406	13,082	14,118	14,531	14,812	15,463	16,283	17,123	17,642	18,668	18,982	18,656	19,145	19,389	20,518	21,261	21,547	23,062	23,448	23,441	24,961	25,534	25,227
E09000030	13,130	13,348	14,069	15,107	15,526	15,546	16,026	16,843	17,850	18,814	19,924	20,184	20,285	20,452	20,344	21,161	21,749	22,095	23,598	24,474	24,602	25,629	26,246	26,175
E09000031	10,594	10,801	11,470	12,209	13,156	13,554	14,193	14,732	15,324	15,720	16,589	16,840	16,809	17,109	17,209	18,394	19,149	19,841	21,504	22,080	22,891	24,304	25,278	25,486
E09000032	13,802	15,350	17,080	19,253	19,833	19,741	20,120	21,666	24,062	26,166	29,249	30,400	30,496	29,325	28,077	29,987	32,826	34,708	36,632	37,589	37,794	39,853	40,665	40,793
E09000033	20,955	21,757	23,748	25,781	26,160	25,842	26,303	28,111	30,396	33,225	37,454	40,370	40,834	40,456	39,513	42,779	46,942	49,486	52,695	53,894	53,729	54,284	52,926	48,382



Let's **combine** them together

Research paper for reference

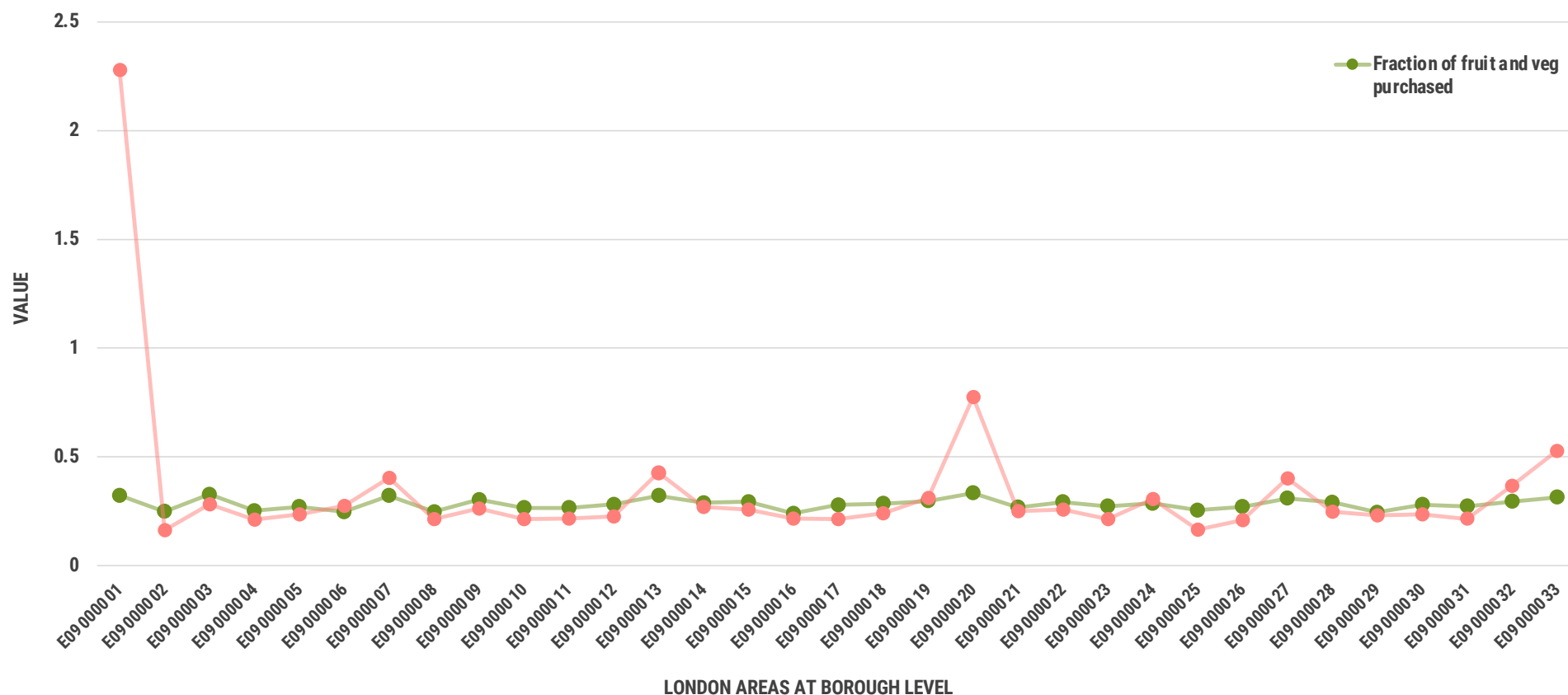
US Adults' Fruit and Vegetable Intakes, 1989 to 1991: A Revised Baseline for the *Healthy People 2000* Objective

*Susan M. Krebs-Smith, PhD, D. Annetta Cook, MS, Amy F. Subar, PhD,
Linda Cleveland, MS, and James Friday*

Finding

Consumption of fruit
and vegetable is
correlated with income.

INSIGHT: What are the differences in fruit and vegetable purchases between areas with high and low gross disposable household income (in £100K)?



Refs: <https://doi.org/10.6084/m9.figshare.c.4769354.v2>

<https://www.ons.gov.uk/economy/regionalaccounts/grossdisposablehouseholdincome/datasets/regionalgrossdisposablehouseholdincomelocalauthoritiesbyitl1region>

References

References

Aiello, L. M., Quercia, D., Schifanella, R., and Prete, L. D., 2020. Tesco Grocery 1.0 [Online], 2. Available from: <https://doi.org/10.6084/m9.figshare.c.4769354.v2> [Accessed 20 March 2023].

Aiello, L. M., Quercia, D., Schifanella, R., and Prete, L. D., 2020. Tesco Grocery 1.0, a large-scale dataset of grocery purchases in London. *Scientific data* [Online], 2. Available from: <https://www.nature.com/articles/s41597-020-0397-7> [Accessed 15 March 2023].

Krebs-Smith, S. M., Cook, D.A., Subar, A. F., Cleveland L., Friday, J., 1995. US Adults' Fruit and Vegetable Intakes, 1989 to 1991: A Revised Baseline for the Healthy People 2000 Objective. *American Journal of Public Health*, 85(12), pp. 1623-1629.

Fenton T., 2022. Regional gross disposable household income: local authorities by ITL1 region [Online]. Available from: <https://www.ons.gov.uk/economy/regionalaccounts/grossdisposablehouseholdincome/datasets/regionalgrossdisposablehouseholdincomelocalauthoritiesbyitl1region> [Accessed 28 March 2023]